

Analyzing Themes and Responses in Spotify User Reviews

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Motivation & Research Questions

Spotify has been the most popular in music streaming, with 252 million premium subscribers as of Q3 2024 (Statista). Beyond tracking subscription numbers, analyzing review content can reveal how platforms manage user satisfaction and what implicit strategies they employ in responding to user concerns.

- How do Spotify users describe their experience in written reviews?
- Which review topics or tones are more likely to receive an official reply—and what does this reveal about platform priorities or transparency?

Updated Methodology

1. **Manual Annotation:** I will hand-label a small, random sample of reviews with two categories.
 - a. Handle misspelled reviews and delete duplicate reviews.
 - b. Predefine 2 categories.
 - i. **App Experience:** Technical feedback on functionality, performance, or interface (technical/UX features)
 - ii. **Other:** General sentiment, ads or pricing complaints, music content (variety of songs/genres, quality of sound, selection of songs/artist/tune; quantity and quality)
 - c. Use a keyword-based function to find a label roughly.
 - d. Select a small subsample to test and manually correct labels used for the prediction task.
2. **Supervised Classification:** Using the final labeled subsample, I will train a supervised classification model (e.g., logistic regression, Lasso, SVM) to assign categories to the full dataset.
 - a. Input features: TF-IDF and Possible improvement: use **word embeddings (Glove)** to enhance classification performance
3. **Sub-topic Exploration:** Within major categories (e.g., “App Experience”), I will explore more themes using unsupervised methods (e.g., topic modeling) to understand what exactly users are complaining about. I will visualize results with word clouds and descriptive summaries to show what drives dissatisfaction or praise under each theme.
4. **Sentiment and Platform Replies:** I will use NRC sentiment scores to measure the emotion each review is. Then, I’ll compare the sentiment of reviews that received a reply versus those that did not.
 - a. This helps answer: Does Spotify reply more to negative reviews or avoid them? Does the platform prefer to respond to polite, neutral feedback? To show whether Spotify has a selective communication style—which may also reflect its priorities or approach to transparency.